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Palaeomagnetism of Lower Cambrian sediments from the Olenek River section (northern Siberia): palaeopoles and the problem of magnetic polarity in the Early Cambrian

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SUMMARY

New palaeomagnetic data from the Lower and Middle Cambrian sedimentary rocks of northern Siberia are presented. During stepwise thermal demagnetization the stable characteristic remanence (ChRM) directions have been isolated for three Cambrian formations. Both polarities have been observed, and mean ChRM directions (for normal polarity) are: Kessyusa Formation (Lower Cambrian) $D=145°$, $I=-40°$, $N=12$, $\alpha_{95}=12.8°$; pole position: $\Phi=38°$S, $\Lambda=165°$E; Erkeket Formation (Lower Cambrian, stratigraphically highly) $D=152°$, $I=-47°$, $N=23$, $\alpha_{95}=6.8°$; pole position: $\Phi=45°$S, $\Lambda=159°$E; Yunkyulyabit-Yuryakh Formation (Middle Cambrian) $D=166°$, $I=-33°$, $N=38$, $\alpha_{95}=4.6°$; pole position: $\Phi=36°$S, L$=140°$E. These poles are in good agreement with the apparent polar wander path based on the bulk of existing Cambrian palaeomagnetic data from the Siberian platform. In Cambrian times, the Siberian platform probably occupied southerly latitudes stretching from about 35° to 0°, and was oriented 'reversely' with respect to its present position. Siberia moved northwards during the Cambrian by about 10° of latitude. This movement was accompanied by anticlockwise rotation of about 30°. The magnetostratigraphic results show the predominance of reversed polarity in the Early Cambrian and an approximately equal occurrence of both polarities in the part of the Middle Cambrian studied. These results are in good agreement with the palaeomagnetic polarity timescale for the Cambrian of the Siberian platform constructed previously by Khramov *et al.* (1987).

Key words: Cambrian, magnetostratigraphy, palaeomagnetic pole, palaeomagnetism, Siberia.

INTRODUCTION

The palaeomagnetic study of the Cambrian sedimentary rocks of the northern part of the Siberian platform was started in the early sixties (see Khramov *et al.* 1974). The Global Palaeomagnetic Database (Lock & McElhinny 1991) contains 56 Cambrian palaeomagnetic poles from Siberia. However, only 11 of them were derived from the northern part of the platform. Only 27 of these poles (seven from the northern part of the platform) were obtained by magnetic cleaning. There are 15 poles for the Middle and Early Cambrian (six from the north). Most of these data were obtained more than 15 years ago, without modern laboratory equipment and such methods as component analysis. Therefore, these data are usually ignored in review papers. The reason is their low 'reliability' according to published reliability criteria (e.g. Briden & Duff 1982; Van der Voo 1990; Pechersky & Didenko 1995). These criteria take account of such factors as the presentation of Zijderveld plots (Zijderveld 1967), details of magnetic cleaning, the number of samples, etc. These factors are of course very important, but they describe mainly the efforts of the author to retrieve palaeomagnetic information, not the palaeomagnetic reliability of the rocks. Consequently, good palaeomagnetic results are sometimes rejected. This is one possible reason for the different interpretations of the palaeomagnetic data from the Cambrian rocks of Siberia. The Late Precambrian–Cambrian apparent polar wander path (APWP) for the Siberian Platform proposed by Pechersky & Didenko (1995) trends from the Gulf of California (Vendian) to southeastern Australia (Middle Cambrian). These authors used only one Lower Cambrian palaeomagnetic pole, that of Kirschvink & Rozanov (1984), because it possessed the highest 'reliability index' among the relevant data. We have constructed a much shorter APWP for this time interval (in the Tasman Sea, see the Discussion section). This path was based on all other data except the pole of Kirschvink & Rozanov (1984). One of the

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reasons for this present study was to obtain new palaeomagnetic data from the Lower Cambrian rocks of Siberia using modern laboratory equipment and methods, and to distinguish between the different interpretations.

There are a number of disagreements in magnetostratigraphic data, especially from the Lower Cambrian rocks of Siberia. For example, the palaeomagnetic scale of Khramov *et al.* (1987) based on the 'old' data shows a strong predominance of reversed polarity in the Early Cambrian. On the other hand, the results of Kirschvink & Rozanov (1984) from carbonate sediments near the Lena River show many polarity reversals in the Early Cambrian. Another aim of the present work was to undertake a detailed magnetostratigraphic study of the whole Lower Cambrian section in Siberia.

GEOLOGY AND SAMPLING

The oldest Cambrian basal beds are exposed on the Olenek uplift in the northeastern part of the Siberian platform. A detailed biostratigraphic description of these exposures has been made by Rozanov *et al.* (1992) and Zinchenko, Vasilieva & Rudavskaya (1993).

One of the most complete sequences of the Lower and Middle Cambrian is exposed along the middle and lower course of the Olenek River (Fig. 1). The Lower Cambrian platform deposits [up to the bottom of the Atdabanian stage (Zinchenko *et al.* 1993)] are represented in these outcrops by the Kessyusa Formation and by the lower part of the Erkeket Formation. The sediments of the Kessyusa Formation overlie the light-coloured dolomites of the Turkut Formation, the boundary being a stratigraphic unconformity.

These Turkut dolomites contain a monotonous complex of stromatolites and microfitolites (Meshkova, Juravleva & Luchinina 1973; Savitsky *et al.* 1972). This complex, according to Rozanov *et al.* (1992) and Zinchenko *et al.* (1993), suggests a Vendian age for the Turkut Formation.

The Kessyusa Formation is divided into three subformations (Vodanyuk & Karlova 1988; Rudavskaya & Vasilieva 1985). The lower subformation consists of three parts: the lower part is composed of sandstones, the middle part is composed of siltstones and the upper part is represented by terrigenous and carbonaceous deposits. The middle subformation is represented mainly by monotonous dark argillaceous sediments. The highly variable upper subformation consists of three parts: the lower part comprises siltstones and sandstones, the middle part is carbonaceous and the upper part is represented by carbonaceous and terrigenous sediments. The whole thickness of the Kessyusa Formation varies from 100 to 110 m.

There are several opinions as to the age of the Kessyusa Formation. According to Rozanov *et al.* (1992) only the upper subformation belongs to the Lower Cambrian (Tommotian stage), and the two lower subformations are of Vendian age (Nemakit-Daldynian regional stage). However, a number of other researchers (e.g. Zinchenko *et al.* 1993) have suggested an Early Cambrian age for the whole Kessyusa Formation based on the occurrence of acritarchs. Vidal, Moczydlowska & Rudavskaya (1995) confirmed an Early Cambrian age for the upper subformation of the Kessyusa Formation. They reported similar acritarch assemblages in the Tommotian stage. There are three K–Ar dates from outcrop 13 (Fig. 1) (Nikolaeva, Zagruzina & Murina 1986), but their range is very large: 461, 505 and 593 Ma. Recently, a new U–Pb zircon age of 543.9 ± 0.24 Ma from the breccia near the base of the Kessyusa Formation on the Khorbosuonka River was obtained (Bowring *et al.* 1993). This result confirms the Early Cambrian age of the Kessyusa Formation.

The Erkeket Formation overlies the Kessyusa Formation with a small erosional gap. It comprises red argillaceous limestones and grey marls. The thickness of the formation varies from 93 to 104 m. According to Zinchenko *et al.* (1993), the Erkeket Formation corresponds to the Tommotian (*regularis* zone), Atdabanian and Lenian (*gurarii* zone) stages of the Lower Cambrian.

Grey beds of the Nouy and Kuonam formations overlie the sediments of the Erkeket Formation. The Kuonam Formation is overlain by mottled limestones and marls of the Yunkyulyabit-Yuryakh Formation. The whole thickness of this formation varies from 75 to 105 m. According to Zinchenko *et al.* (1993) the Yunkyulyabit-Yuryakh Formation belongs to the Amginian (*fissus-sacheri* zone) and Mayan (*limbataeformis* zone) stages of the Middle Cambrian [Solvan and Menevian stages respectively, according to Harland *et al.* (1989)]. All the formations described are present in the outcrops shown in Fig. 1. The strata in the study area are essentially flat-lying with dips usually less than 2°.

A total of 381 oriented hand-samples were collected from the outcrops shown in Fig. 1. Among them are 59 samples from the Kessyusa Formation (outcrops 2, 8, 11 and 13), 209 samples from the Erkeket Formation (outcrops 1, 7, 8, 10, 11, 12, 13 and 14), 92 samples from the Yunkyulyabit-Yuryakh Formation (outcrops 5, 6, 14 and 15) and 21 samples from the Turkut Formation (outcrops 2 and 9). The sampling density at each outcrop was usually one sample per 1–2 m of thickness, with a sampling interval from 8 to 150 m. The sampling extended over a distance of 200 km. A magnetic compass was used for orientation. The positions of outcrops with respect to the stratigraphic column are shown in Fig. 1.

PALAEOMAGNETIC ANALYSIS

From two to six cubic specimens with sides of 2 and 2.4 cm were made from each sample. The natural remanent magnetization (NRM) was measured on the SQUID magnetometer in the Geophysical Laboratory of Aarhus University (Denmark), the Czech spinner-magnetometer JR4 and on the astatic magnetometer VNIGRI (VNIGRI, St Petersburg, Russia). Thermal demagnetization in non-magnetic space (the residual field was less than 4–10 nT) at temperatures of 100, 200, 250, 300, 350, 400, 450 and 500 °C was used for retrieving the characteristic remanent magnetization (ChRM) from almost all specimens. Some specimens were also heated further to temperatures of 550, 570, 600, 620, 660, 670 and 690 °C. Some of the other specimens were also subjected to alternating magnetic field (AF) demagnetization in steps of 10 mT to a maximum field of 100 mT, but this type of magnetic cleaning was ineffective for the rocks studied. Thermal demagnetization was carried out using a Schonstedt TSD-1 demagnetizer, and AF demagnetization was achieved using a Molspin demagnetizer (Geophysical Laboratory of Aarhus University). The magnetic susceptibility was measured after each heating step with a Molspin kappameter (at Aarhus University) and Geofyzika KLY-2 bridge (at VNIGRI). The 'compensation method' (Khramov & Sholpo 1967) was used for the estimation of the percentage of viscous remanent magnetization (VRM) in the

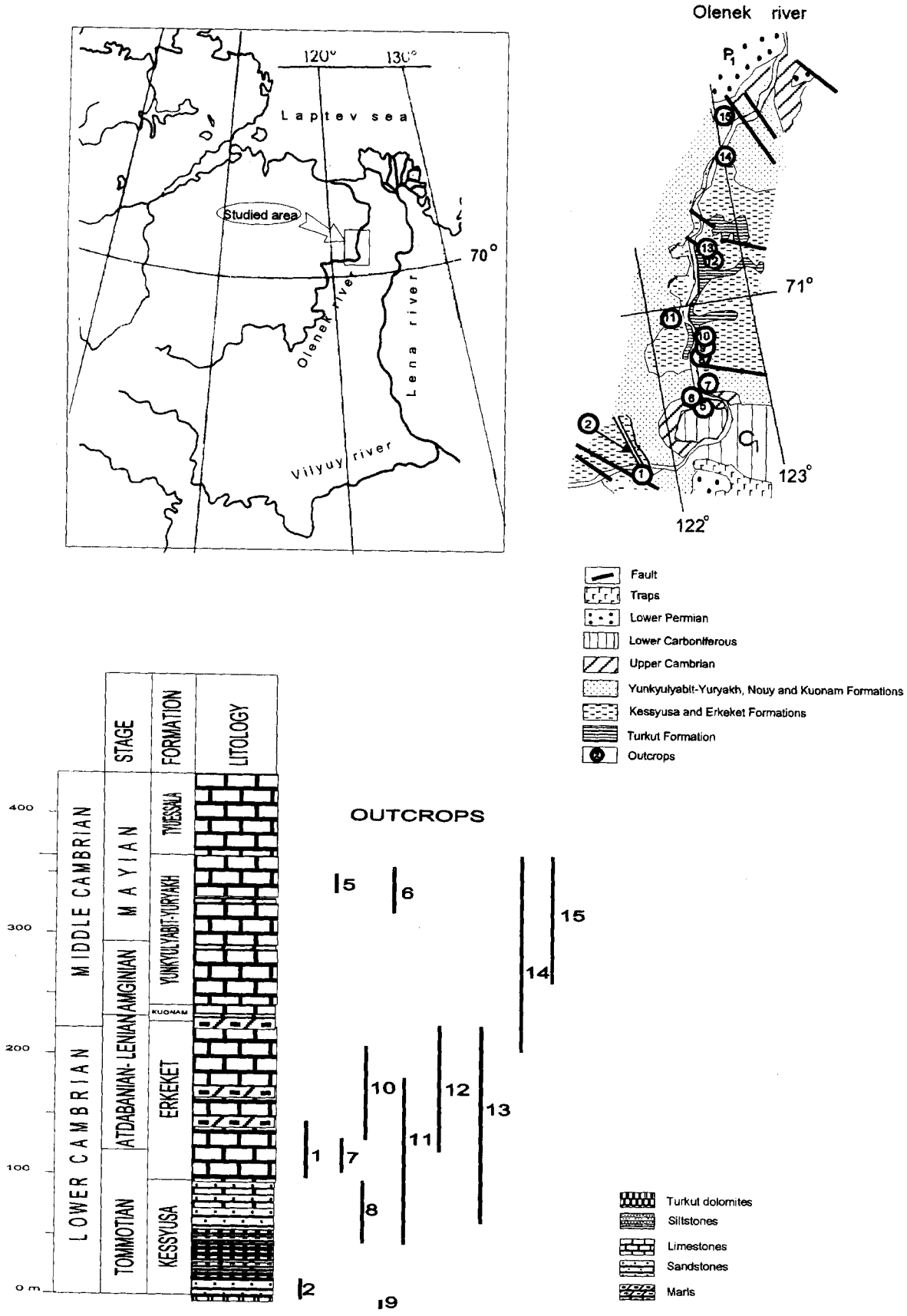


Figure 1. Sketch map of the area sampled and stratigraphic positions of the outcrops studied.

NRM. The specimens were put into *in situ* position for 1–2 months and then, after measuring, into the opposite position for 4–6 months. If the NRM changed by more than 40 per cent, the specimen was excluded from the further study. The IAPD computer program of T. Torsvik was used for data analysis [the principal component analysis technique of Kirschvink (1980) in particular]. Isothermal remanent magnetization (IRM) experiments were carried out in the Laboratory of Rock Magnetism (St Petersburg University).

It was impossible to carry out a fold test due to the horizontal bedding, and the conglomerate test was also impossible due to the absence of convenient rocks. Only the reversal test (McFadden & McElhinny 1990) has been carried out. Fisher (1953) statistics were used for formation mean calculations.

Turkut Formation

Light-coloured dolomites of the Turkut Formation are very weakly magnetized. NRM is less than 5×10^{-3} mA m^{-1} and its direction is very close to that of the present-day geomagnetic field.

Kessyusa Formation

The NRM of these sediments ranges from 1 to 40 mA m^{-1}, and the magnetic susceptibility from 0.06 to 2×10^{-3} SI units. Unfortunately, most of the NRM is carried by unstable minerals: the magnetic susceptibility in many cases increased by up to 10 times after heating to 300 °C (and to lower temperatures in some specimens). The direction of the low-temperature component is close to the direction of the present-day geomagnetic field (Fig. 6d). Nevertheless, some specimens show the presence of a high-temperature component. This component is probably connected with fine magnetite particles as well as with haematite: a significant part of the NRM remains even after 600 °C (Fig. 3). IRM curves (Fig. 2a) saturate in a magnetic field of about 0.6 T. This, together with a relatively low value of the coercivity of remanence (50 mT), suggests a dominance of magnetite in these rocks. The direction of the high-temperature component of remanence was determined with confidence only for 12 samples (16 specimens): an example is shown in Fig. 3. Confident interpretation was impossible in other cases due to the random behaviour of the magnetization during cleaning. The stereoplot for this component is shown in Fig. 6(a), and the correspondent values and statistics in Table 1 (line 1). This mean direction is based on samples from all parts of the Kessyusa Formation: the small number of determinations prevents us from presenting separate means for the different parts of the formation. For the same reason we did not try to make a magnetostratigraphic column for the Kessyusa Formation. However, we think that these results suggest a dominance of reversed polarity in the rocks studied (directions with a positive inclination according to the APWP for Siberia, see below). The directions and statistics for the groups of normal and reversed polarity are, respectively, $N=2$, $D=165.7°$, $I=-40.2°$ and $N=10$, $D=323.3°$, $I=39.7°$, $k=10.9$, $\alpha_{95}=15.3°$. The reversal test (McFadden & McElhinny 1990) is indeterminate due to a large scatter.

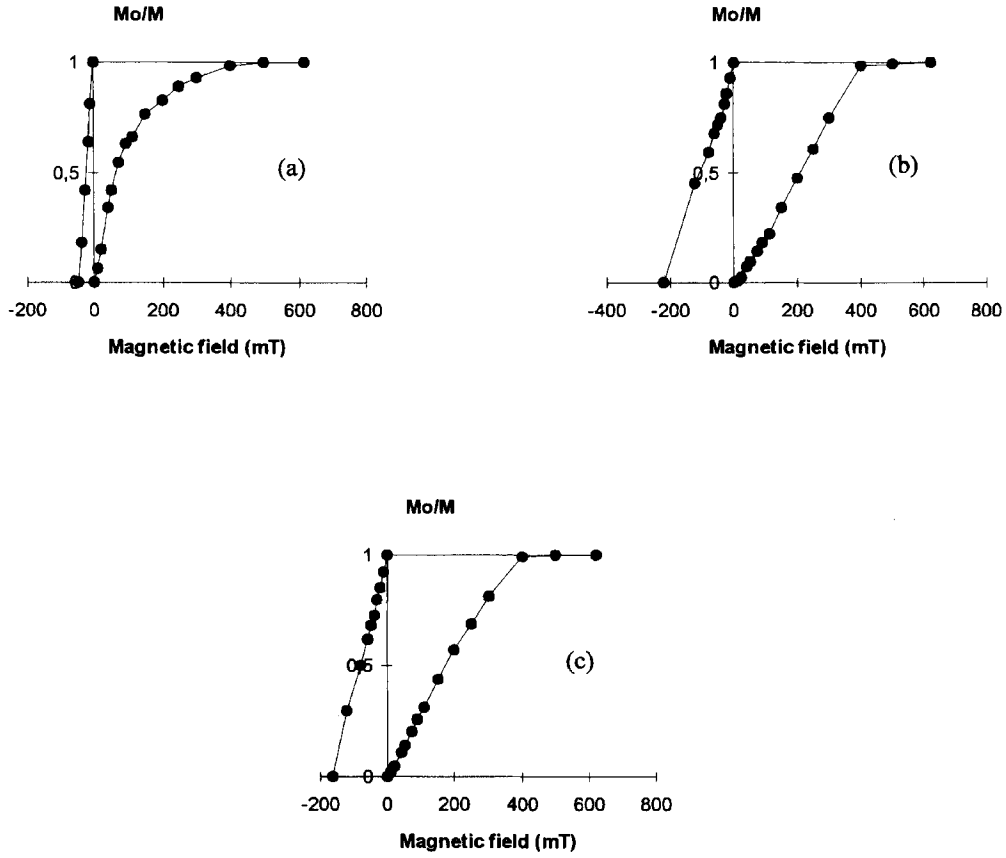


Figure 2. Examples of IRM curves. Samples from (a) the Kessyusa Formation (Early Cambrian); (b) the Erkeket Formation (Early Cambrian); (c) the Yunkyulyabit-Yuryakh Formation (Middle Cambrian).

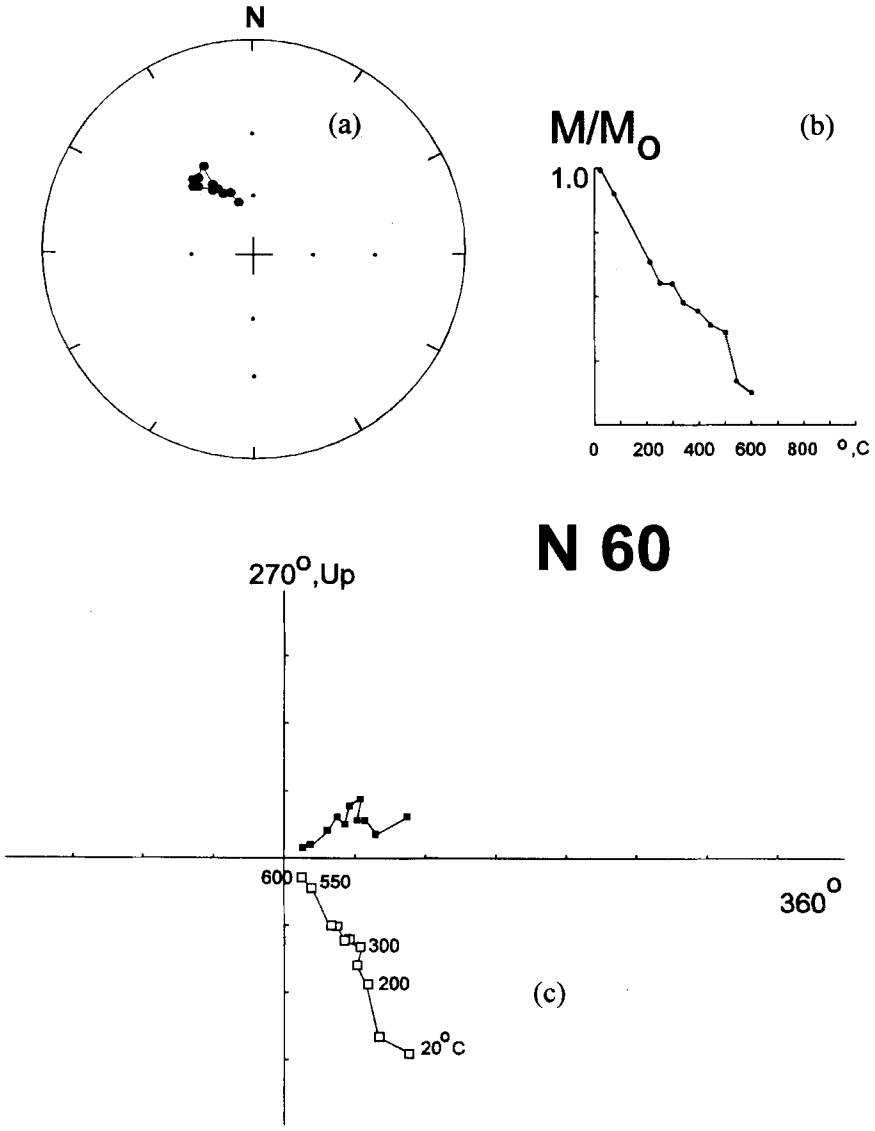


Figure 3. Example of thermal demagnetization of the Early Cambrian sediments of the Kessyusa Formation (sample number 60). (a) Stereographic projection of ChRM directions; solid and open symbols represent downward and upward directions, respectively. (b) Demagnetization curve. (c) Zijderveld (1967) diagram; solid and open symbols denote horizontal and vertical projections, respectively.

Table 1. Mean palaeomagnetic directions and pole positions.

	Formation name	*B*/*N*/*n*	*D* (°)	*I* (°)	*k*	α_{95} (°)	Φ (°N)	Λ (°E)	*dp* (°)	*dm* (°)
1	Kessuysa	4/12/16	144.6	−39.8	10.0	12.8	−37.6	165.0	9.3	15.4
2	Erkeket	8/23/31	151.5	−47.4	20.5	6.8	−44.8	158.7	5.7	8.8
3	Yunkyulyabit-Yuryakh	3/38/76	165.6	−33.2	26.4	4.6	−36.4	139.6	3.0	5.2

B = number of outcrops, *N* = number of samples, *n* = number of specimens; *D*, *I* = declination and inclination of ChRM; *k* = precision parameter (Fisher 1953); α_{95} = 95 per cent confidence circle; Φ, Λ = latitude and longitude of VGP; *dp*, *dm* = semi-axis of the oval of 95 per cent confidence about the mean pole. All statistics are on the sample level.

Erkeket Formation

The NRM of the red limestones of the Erkeket Formation ranges from 0.3 to 5.1 mA m^{-1} and the magnetic susceptibility from 0.02 to 0.06 × 10^{-3} SI units. The magnetic susceptibility of these rocks did not change during heating up to 670 °C. Almost all specimens possess a low-temperature NRM component (T_{ub} = 300–400 °C). Its direction is close to the direction of the present-day geomagnetic field (Fig. 6d). Unfortunately, this component has been the only one determined in many samples: the component remaining after heating to higher temperatures has been at the noise level of the demagnetizing equipment.

However, a high-temperature component ($T_{ub}>400\,°C$) has been retrieved in 25 samples (35 specimens). Magnetite is probably the carrier of this component: the magnetization in many cases decreases almost to zero after heating to 500 °C (Fig. 4). IRM curves (Fig. 2b) are caused mainly by magnetite. However, the less steep form of the curve, the higher value of the coercivity of remanence (220 mT) and the red colour of the rock lead us to conclude that haematite is present. The size of haematitic particles is probably very small, close to the critical size of the superparamagnetic state. This suggestion can explain the visible absence of a high-temperature (haematite) component of the NRM (Fig. 4). The stereoplot is shown in Fig. 6(b), and the mean direction and statistics are presented in Table 1 (line 2). As in the case of the Kessyusa Formation, reversed polarity predominates over normal. Mean directions and statistics for the groups of normal and reversed polarities are, respectively, $N=3$, $D=162.8°$, $I=-39.6°$, $k=54.1$, $\alpha_{95}=16.9°$ and $N=20$, $D=329.6°$, $I=48.5°$, $k=19.7$, $\alpha_{95}=7.5°$. The reversal test gave a positive result, R_c, according to McFadden & McElhinny (1990).

Yunkyulyabit-Yuryakh Formation

The best palaeomagnetic results in this study were obtained from the rocks of the Yunkyulyabit-Yuryakh Formation. Their NRM and magnetic susceptibility are of the same range as those of the Erkeket Formation: from 0.2 to 6.6 mA m^{-1} and from 0.06 to 0.13×10^{-3} SI units, respectively. However, the percentage of low-temperature (probably VRM) component in these rocks is much less. An example of the thermal demagnetization is shown in Fig. 5. The viscous component was usually removed after heating to 200 °C, after which the direction of magnetization during further heating (up to 660 °C) did not change significantly. The intensity of magnetization

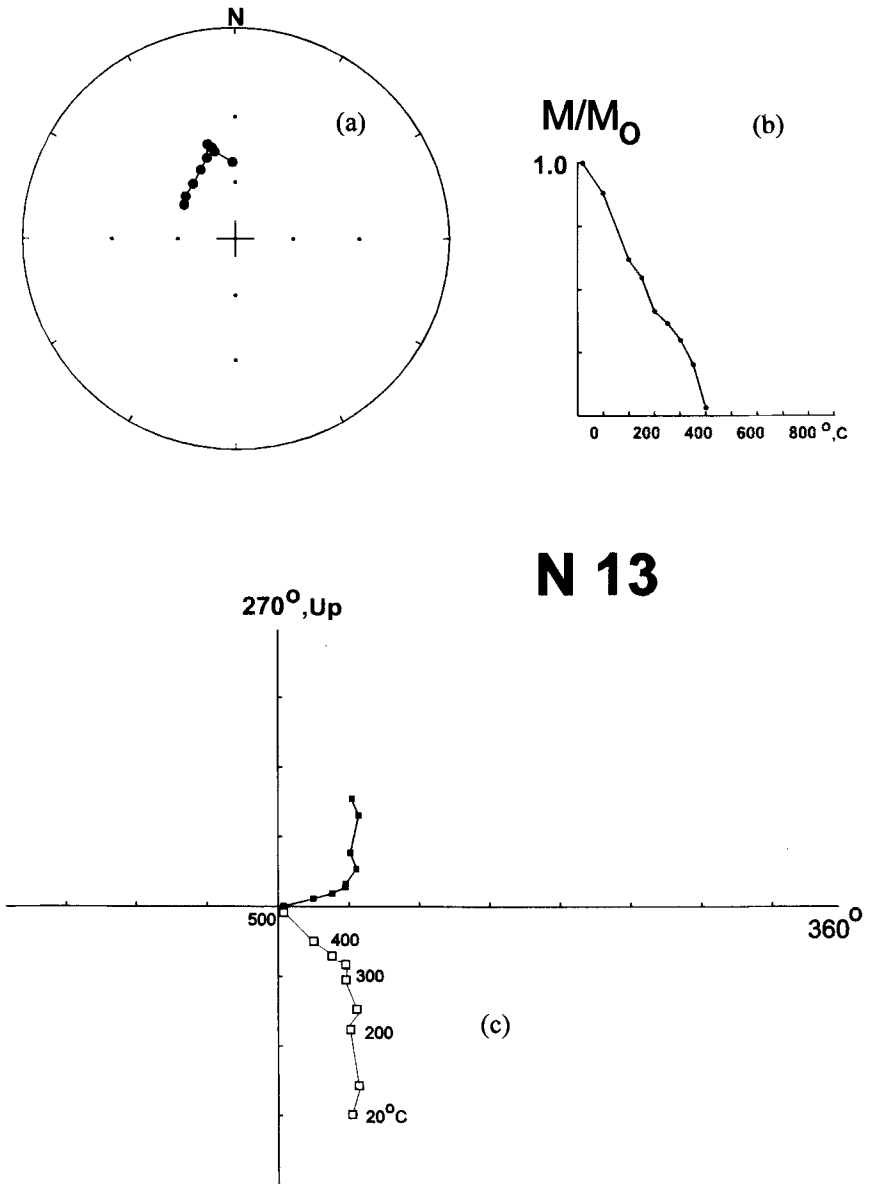


Figure 4. Example of thermal demagnetization of the Early Cambrian sediments of the Erkeket Formation (sample number 13). Notation as for Fig. 3.

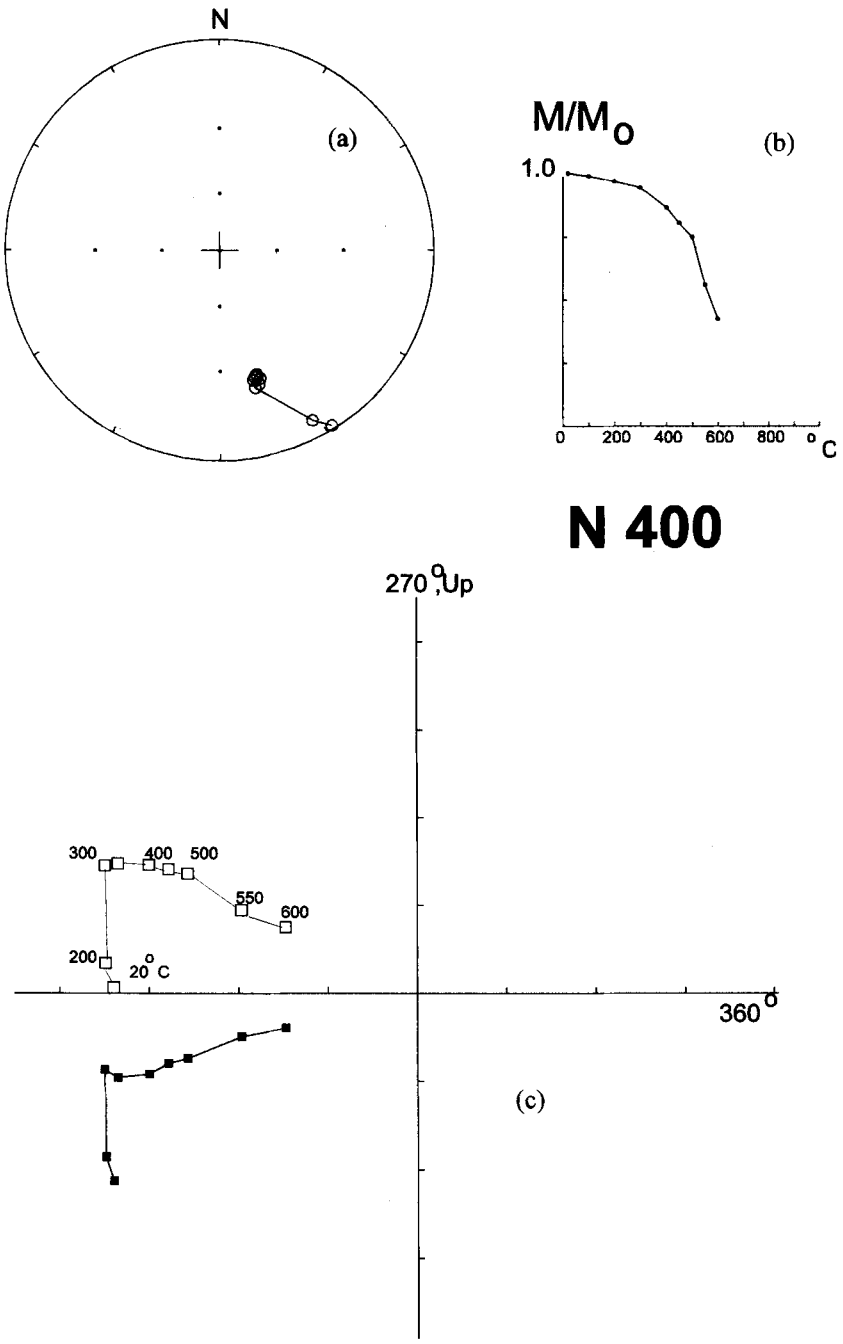


Figure 5. Example of thermal demagnetizion of the Middle Cambrian sediments of the Yunkyulyabit-Yuryakh Formation (sample number 400). Notation as for Fig. 3.

decreased slowly and even after 610 °C remained at about a half of its NRM value. Such behaviour suggests haematite as the main carrier of ChRM. IRM curves (Fig. 2c) and the red colour of the rocks support this suggestion. The results from 16 samples were rejected due to a large within-sample scatter. The results are plotted on a stereoplot in Fig. 6(c), and the mean direction and statistics are given in Table 1 (line 3). Both polarities are present almost equally in the section; mean directions and statistics for normal and reversed polarities are, respectively, $N=17$, $D=164.3°$, $I=-35.6°$, $k=39.1$, $\alpha_{95}=5.4°$ and $N=20$, $D=346.9°$, $I=31.0°$, $k=19.9$, $\alpha_{95}=7.5°$. The dual-polarity magnetizations are approximately reversed with respect to each other and yield a positive reversals test with an R_b classification (McFadden & McElhinny 1990).

DISCUSSION

As mentioned in the Introduction, there are 27 Cambrian palaeomagnetic results from the Siberian platform with a DEMAGCODE>0 (Lock & McElhinny 1991); these are summarized in Table 2. It is clear from this table that the great majority of palaeomagnetic poles are situated close to each other, and the Early Cambrian poles are situated slightly to the east of the Middle and Late Cambrian poles. The exception

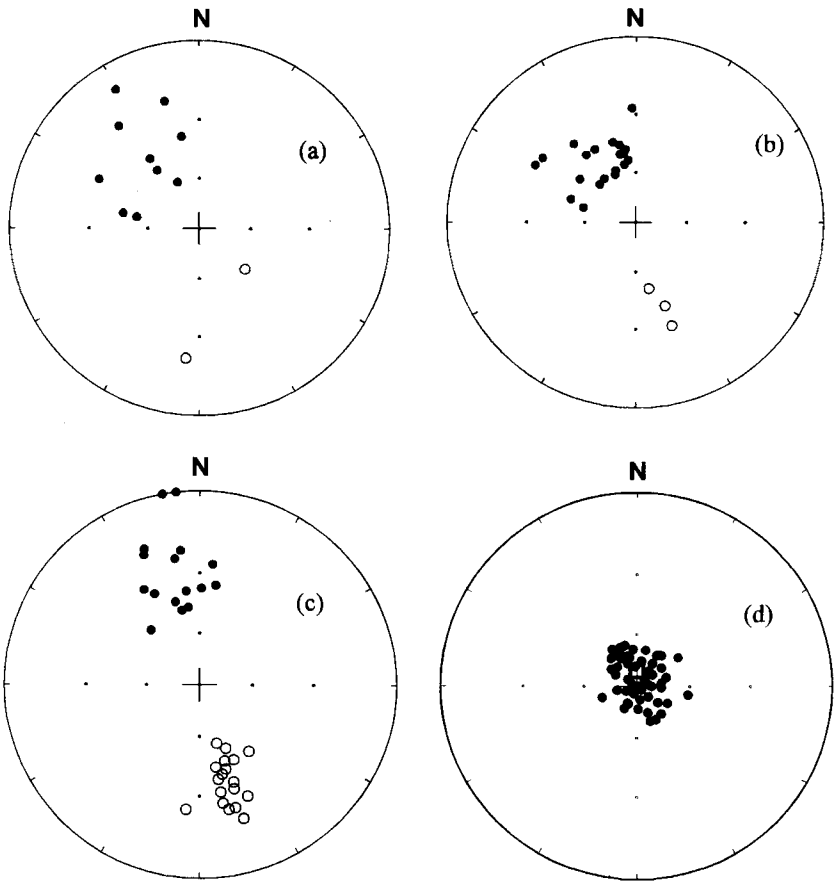


Figure 6. Stereoplots of sample mean palaeomagnetic directions: (a) Kessyusa Formation (Early Cambrian); (b) Erkeket Formation (Early Cambrian); (c) Yunkyulyabit-Yuryakh Formation (Middle Cambrian); (d) low-temperature component in all formations studied (the square represents the present local field direction).

is the result of Kirschvink & Rozanov (1984). As some APWPs based on this result have already been constructed (Pechersky & Didenko 1995; Evans *et al.* 1996) and discussed, we decided to create an alternative model using all the results that we believe are reliable, except that mentioned above.

Using the GMAP program of T. Torsvik and M. Smethurst, with the algorithm of Jupp & Kent (1987), we have constructed a Cambrian fragment of the APWP for the Siberian platform using data from Table 2, as shown in Fig. 7. New palaeomagnetic poles from Table 1 and their confidence ovals are also shown in this diagram. They are in a good agreement with this APWP. Ovals of Early Cambrian poles overlap, but these poles are significantly different from the Middle Cambrian pole.

Fig. 8 shows three successive palaeolatitudinal and azimuthal positions of the Siberian platform for the Early, Middle and Late Cambrian. These positions were calculated using GMAP and the results of this study, together with data from Table 2 (for the Late Cambrian). During Cambrian times Siberia occupied low southerly latitudes and had an inverted orientation with respect to its present position. During this period there was a relatively short northward latitudinal displacement (about 10°) of the plate accompanied by anticlockwise rotation of about 30°. These positions and movements are in reasonable agreement with various Palaeozoic global reconstructions (e.g. Li, Powell & Trench 1993; Golonka, Ross & Scotese 1994). This finding could be an indication of the reliability of the 'old' palaeomagnetic data from the Siberian platform, in spite of their low 'reliability index'.

Of course, the discrepancy between our Early Cambrian poles and the pole of Kirschvink & Rozanov (1984) cannot be explained completely. We cannot dismiss entirely the possibility of Middle Cambrian remagnetization; however, it seems unlikely (the significant difference between the Middle Cambrian and Early Cambrian directions, two polarities, the absence of geological evidence of regional heating, etc.). Another possibility is the proposed pre-Devonian opening of the Viljui depression, which led to a 30° clockwise rotation of the Aldan Shield with respect to the northern Siberian plate (Gurevich 1984), but even in this case the difference between our pole position and that from Kirschvink & Rozanov (1984) revised by Kirschvink *et al.* (1991) will be about 60° (though not 90°).

The combined magnetostratigraphic section of the formations studied is shown in Fig. 9. As mentioned above, we did not build a detailed magnetostratigraphic column for the Lower Cambrian part of the section studied and made a conclusion only about the dominance of reversed polarity. This part of the section is thus presented in Fig. 9 as a zone of 'predominantly reversed polarity'. However, we have shown the stratigraphic positions of few normally magnetized samples. An additional study is necessary to build a detailed magnetostratigraphic column of the Lower Cambrian of Siberia.

Table 2. Palaeopoles for Cambrian rocks of the Siberian Platform and corresponding sampling locations.

N	Age	Location		Pole		Reference
		ϕ (°N)	λ (°E)	Φ (°N)	Λ (°E)	
5410	L. Cam.	59.5	112.5	−33.0	121.0	Davydov, V.F. & Kravchinsky, A.Y. in Khramov 1973
5384	L. Cam.	58.0	97.0	−37.0	140.0	Gurariy, G.Z. in Khramov 1973
5390	L. Cam.	54.3	104.6	−34.0	134.0	Gurariy, G.Z. & Trubikhin, V.M. in Khramov 1973
5391	L. Cam.	57.0	107.0	−39.0	125.0	Rodionov, V.P. in Khramov 1973
5406	L. Cam.	67.5	110.5	−35.0	130.0	Rodionov, V.P. in Khramov 1973
5407	L. Cam.	60.0	114.0	−34.0	103.0	Rodionov, V.P., Osipova, E.P. in Khramov 1971
5411	L. Cam.	58.0	109.0	−39.0	129.0	Rodionov, V.P. in Khramov 1975
5416	L. Cam.	58.3	109.7	−38.0	125.0	Rodionov, V.P. in Khramov 1973
5437	L. Cam.	55.0	105.0	−29.0	129.0	Kravchinsky, A.Y. & Davydov, V.F. in Khramov 1979
5438	L. Cam.	58.7	111.0	−30.0	119.0	Rodionov, V.P. in Khramov 1982
5439	L. Cam.	59.0	112.0	−40.0	114.0	Rodionov, V.P. in Khramov 1982
5445	L. Cam.	54.5	105.5	−32.0	133.0	Rodionov, V.P. in Khramov 1982
5446	L. Cam.	58.0	108.8	−36.0	126.0	Rodionov, V.P. in Khramov 1982
5447	L. Cam.	58.6	110.3	−34.0	120.0	Rodionov, V.P. in Khramov 1982
5403	L. Cam.	54.5	105.5	−36.0	134.0	Kravchinsky, A.Y. & Davydov, V.F. in Khramov 1989
5424	M. Cam.	61.5	135.5	−45.0	164.0	Osipova, E.P. in Khramov 1975
5395	M. Cam.	68.0	112.0	−26.0	134.0	Osipova, E.P. in Khramov 1971
5420	M. Cam.	68.0	112.0	−38.0	141.0	Osipova, E.P. in Khramov 1973
5423	M. Cam.	60.0	133.0	−45.0	154.0	Osipova, E.P. in Khramov 1973
5448	M. Cam.	68.0	112.0	−31.0	141.0	Rodionov, V.P. in Khramov 1982
5474	M. Cam.	71.0	128.0	−41.3	167.4	Osipova, E.P. in Khramov 1989
5409	E. Cam.	55.5	97.5	−44.0	140.0	Davydov, V.F. & Kravchinsky, A.Y. in Khramov 1971
5471	E. Cam.	71.5	116.0	32.0	137.0	Rodionov, V.P. in Khramov 1986
1131	E. Cam.	61.0	126.8	16.6	244.5	Kirschvink & Rozanov 1984
5428	E. Cam.	68.5	112.5	−39.0	153.0	Osipova, E.P. in Khramov 1975
5429	E. Cam.	59.5	135.0	−46.0	181.0	Osipova, E.P. in Khramov 1975
5470	E. Cam..	59.0	135.0	−46.0	162.0	Osipova, E.P. in Khramov 1986

N: result number in the Global Palaeomagnetic Database (Lock & McElhinny 1991); ϕ, λ: latitude and longitude of the sampling area; Φ, Λ: latitude and longitude of the palaeopole.

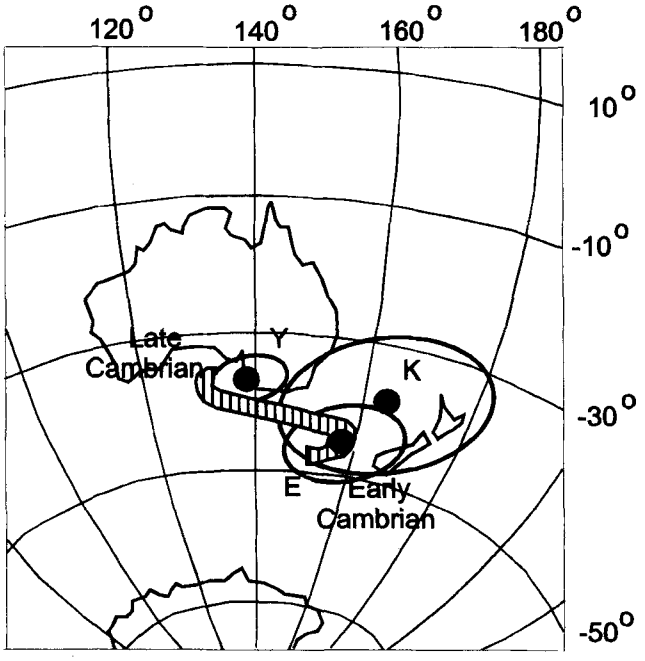


Figure 7. The Cambrian fragment of the APWP for Siberia constructed using the results from Table 2, and positions of palaeomagnetic poles from this study with ovals of confidence. K: Kessyusa Formation; E: Erkeket Formation; Y: Yunkyulyabit-Yuryakh Formation.

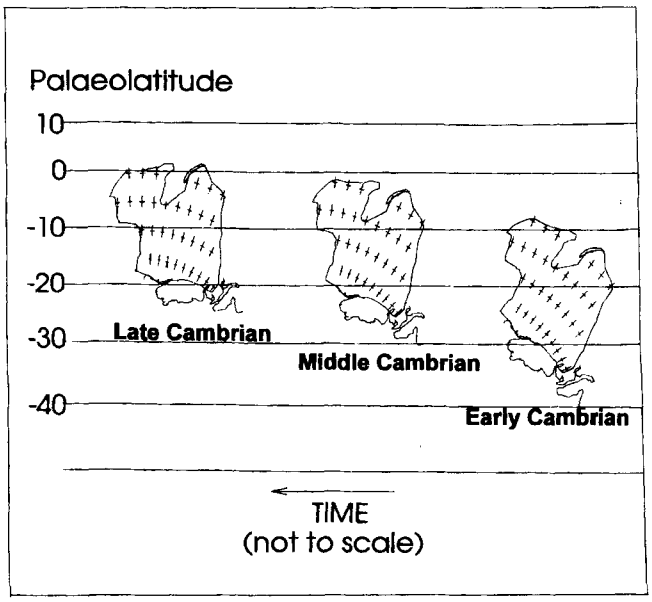


Figure 8. Palaeolatitudinal and azimuthal positions of the Siberian Platform during the Cambrian.

Analysis of the Global Palaeomagnetic Database (Lock & McElhinny 1991) gives us indirect confirmation of the conclusion about the predominance of reversed polarity in the Early Cambrian. There are 45 palaeomagnetic results from Lower Cambrian rocks all over the world. 27 of them show

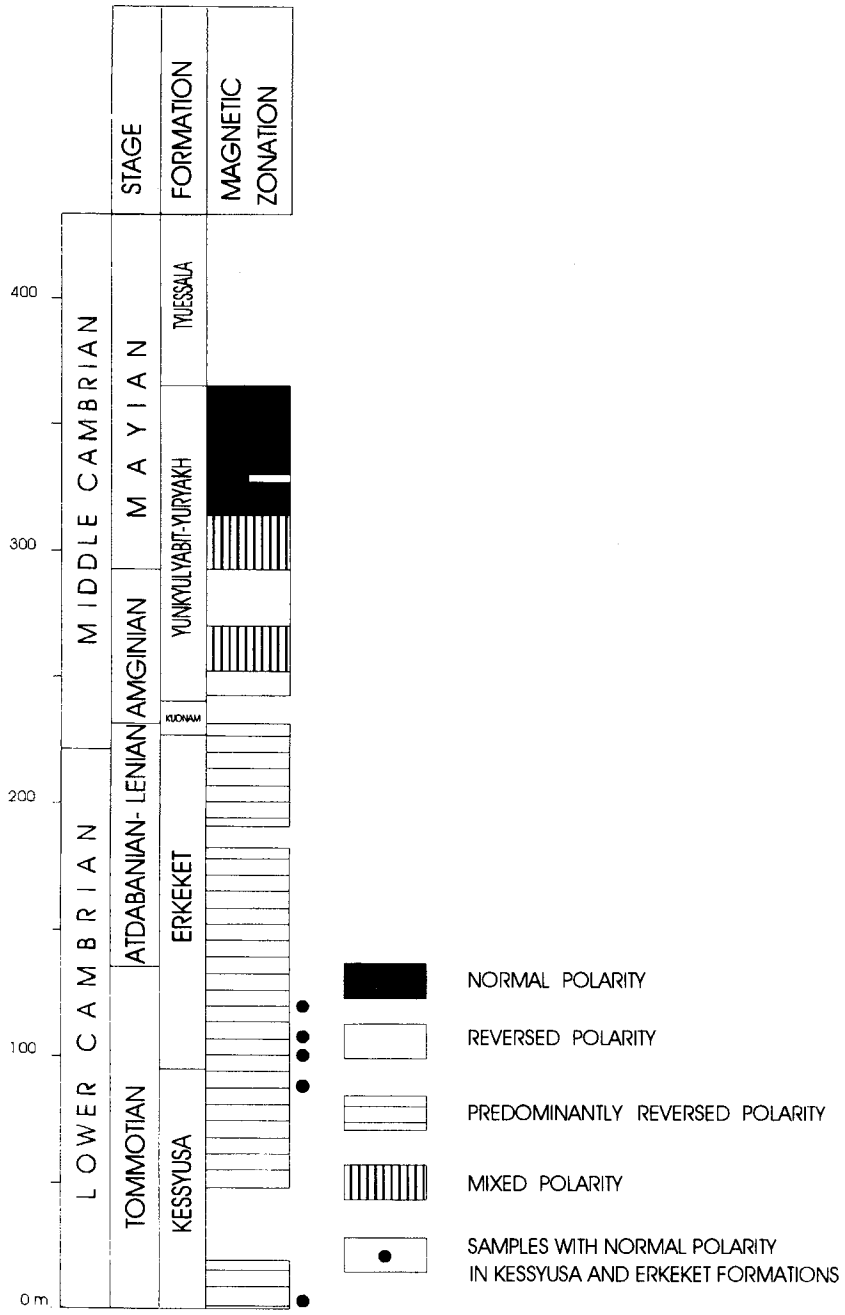


Figure 9. The combined magnetostratigraphic section of the formations studied in this work.

the presence of reversed polarity only, 14 contain both polarities, and only four show the presence of exclusively normal polarity. The principal magnetostratigraphic section of the Lower Cambrian of Siberia (Khramov *et al.* 1987) also shows a dominance of reversed polarity. Similar results have been determined from Pakistan (Klootwijk, Nazirullah & DeJong 1986) and from Northern China (Zhao *et al.* 1992). Algeo (1996) reported mixed polarity in the Early Cambrian, but this conclusion is based mainly on the results of Kirschvink & Rozanov (1984), which have recently been revised by the authors themselves (Kirschvink *et al.* 1991).

According to our data, polarity was also reversed in the beginning of the Middle Cambrian. In the lower part of the Amginian stage (Slovanian, according to Harland *et al.* 1989) both polarities are present equally, and the dominance of reversed polarity was established at the end of this stage. At the beginning of the Mayan stage (Menevian, according to Harland *et al.* 1989) there were many geomagnetic reversals, and a stable normal-polarity epoch began later. This sequence is in a good agreement with previous results from the Middle Cambrian rocks of Siberia (Khramov *et al.* 1987).

CONCLUSIONS

New palaeomagnetic data from Lower and Middle Cambrian rocks from the northern part of the Siberian platform have improved the palaeomagnetic Cambrian database. The new palaeomagnetic poles are in good agreement with the APWP

of the Siberian plate based on a majority of the available palaeomagnetic data. However, they are discordant with the result of Kirschvink & Rozanov (1984) from the Lena River section. Additional study is necessary to explain the reason for this discordance. The palaeolatitudinal and azimuthal positions of the Siberian platform for the Cambrian constructed on the basis of the new data confirm several global reconstructions for this time. The new palaeomagnetic results show the dominance of reversed geomagnetic polarity in the Early Cambrian, although it seems that there were about 10 geomagnetic reversals during that time (Fig. 9). The combined magnetostratigraphic column for the Middle Cambrian (Yunkyulyabit-Yuryakh Formation) shows a mixed-polarity zone in the middle of the Amginian (Slovanian) stage, a reversed zone at the top of this stage, a mixed-polarity zone in the lower part of the Mayan (Menevian) stage and a normal zone in the upper part of this stage. Although these magnetostratigraphic results are in reasonable agreement with a majority of previous ones, the necessity of a detailed magnetostratigraphic investigation of the Siberian Lower Cambrian is clear.

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REFERENCES

Algeo, T.J., 1996. Geomagnetic polarity bias patterns through the Phanerozoic, *J. geophys. Res.*, **101,** B2, 2785–2814.

Bowring, S.A., Grotzinger, J.P., Isachsen, C.E., Knoll, A.H., Pelechaty, S.M. & Kolosov, P., 1993. Calibrating rates of early Cambrian evolution, *Science,* **261,** 1293–1298.

Briden, J.C. & Duff, B.A., 1982. Pre-Carboniferous paleomagnetism of Europe north of the Alpine orogenic belt, in *Paleoreconstruction of the Continents,* eds McElhinny, M.W. & Valencio, D.A, *Geodynamics Series,* **2,** 137–150.

Evans, D.A., Zhuravlev, A.Yu., Budney, C.J. & Kirschvink, J.L., 1996. Palaeomagnetism of the Bayan Gol Formation, western Mongolia, *Geol. Mag.,* **133,** 487–496.

Fisher, R.A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.,* A, **217,** 295–305.

Golonka, J., Ross, M.I. & Scotese, C.R., 1994. Phanerozoic paleogeographic and paleoclimatic maps, in *Pangea: Global Environments and Resources,* pp. 1–48, eds Embry, A.F., Beauchamp, B. & Glass, D.J., Canad. Soc. Petrol. Geol. Memoir 17, Calgary, Alberta, Canada.

Gurevich, E.L., 1984. Paleomagnetism of the Ordovician deposits of the Moyero river sequence, in *Paleomagnetic Methods in Stratigraphy,* pp. 35–41, VNIGRI, St Petersburg, Russia (in Russian).

Harland, W.B., Armstrong, R.L., Cox, A.V., Craig, L.E., Smith, A.G. & Smith, D.G., 1989. *A Geologic Time-Scale,* Cambridge University Press, Cambridge.

Jupp, P.E. & Kent, J.T., 1987. Fitting smooth paths to spherical data, *Appl. Stat.,* **36,** 34–36.

Khramov, A.N. (ed). *Palaeomagnetic directions and palaeomagnetic poles. Reference book for the USSR. Materials of the WDC-B, Moscow* (issue 1–1971, issue 2–1973, issue 3 – 1975, issue 4–1979, issue 5–1982, issue 6–1986, issue 7–1989) (in Russian).

Khramov, A.N. *et al.* 1974. *Palaeomagnetism of Palaeozoic,* Nedra, Leningrad (in Russian).

Khramov, A.N. *et al.* 1987. *Paleomagnetology,* Springer–Verlag, Heidelberg.

Khramov, A.N. & Sholpo, L.E., 1967. *Palaeomagnetism. Principles, methods and geological applications of palaeomagnetology,* Nedra, Leningrad (in Russian).

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. astr. Soc.,* **62,** 699–718.

Kirschvink, J.L., Magaritz, M., Ripperdan, R.L., Zhuravlev, A.Yu. & Rozanov, A.Yu., 1991 Precambrian/Cambrian boundary: magnetostratigraphy and carbon isotopes resolve correlation problems between Siberia, Morocco and South China, *GSA Today,* **1,** 69–91.

Kirschvink, J.L. & Rozanov, A.Yu., 1984. Magnetostratigraphy of lower Cambrian strata from the Siberian Platform: a palaeomagnetic pole and a preliminary polarity time-scale, *Geol. Mag.,* **121,** 189–203.

Klootwijk, C.T., Nazirullah, R. & DeJong, K.A., 1986. Paleomagnetic constraints on formation of the Mianwali reentrant, Trans-Indus and western Salt Range, Pakistan, *Earth planet. Sci. Lett.,* **80,** 394–414.

Li, Z.X., Powell, C.McA. & Trench, A., 1993. Palaeozoic global reconstructions, in: *Palaeozoic Vertebrate Biostratigraphy and Biogeography,* pp. 25–53, ed. Long, J.A., Belhaven Press, London.

Lock, J. & McElhinny, M.W., 1991. The Global Paleomagnetic Database, *Surv. Geophys.,* **12,** 1–492.

McFadden, P.L. & McElhinny, M.W., 1990. Classification of the reversal test in paleomagnetism, *Geophys. J. Int.,* **103,** 725–729.

Meshkova, N.P., Juravleva, I.T. & Luchinina, V.A., 1973. Lower Cambrian and the lower part of Middle Cambrian of the Olenek uplift, in *Problemi paleontologii i biostratigrafii nizhnego kembriya Sibiri i Dalnego Vostoka,* pp. 194–214, ed. Juravleva, I.T., Novosibirsk, Nauka (in Russian).

Nikolaeva, I.B., Zagruzina, I.A. & Murina, G.A., 1986. Geochronological boundaries of the Phanerozoic, in *Metodi izotopnoi geologii i geokhronologicheskaya shkala,* pp. 183–215, eds Shukolukov, Ju.A. & Bibikova, E.V., Nauka, Moscow (in Russian).

Pechersky, D.M. & Didenko, A.N., 1995. Paleoasian Ocean: petromagnetic and paleomagnetic information about its lithosphere, OIFZ RAN, Moscow (in Russian).

Rozanov, A.Yu. *et al.* 1992. *Cambrian of Siberia,* Novosibirsk, Nauka (in Russian).

Rudavskaya, V.A. & Vasilieva, N.I., 1985. Acritarchs and skeleton problems on the Vendian, Tommotian and Atdabanian boundaries, in *Stratigrafiya pozdnego dokembriya i rannego paleozoya Sibirskoi platformy,* pp. 51–57, VNIGRI, Leningrad (in Russian).

Savitsky, V.E., Shabanov, Yu.J., Evtushenko, V.H. & Shishkin, B.B., 1972. The principal section of the Cambrian of the Khorbosuonka facial area (north-eastern part of the Siberian platform), in *Materiali po stratigrafii i paleontologii Sibirskoi platformi,* Novosibirsk, SNIIGGIMS, **146,** 53–60 (in Russian).

Van der Voo, R., 1990. Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions, *Rev. Geophys.,* **28,** 167–206.

Vidal, G., Moczydlowska, M. & Rudavskaya, V.A., 1995. Constraints on the early Cambrian radiation and correlation of the Tommotian and Nemakit-Daldynian regional stages of eastern Siberia, *J. Geol. Soc. Lond.,* **152,** 499–510.

Vodanyuk, S.A. & Karlova, G.A., 1988. About the Kessusinskaya suite of the Olenek uplift, in *Pozdnii dokembrii i rannii paleozoi Sibiri. Rifei i Vend,* pp. 3–20, ed. Khomentovski, V.V., Novosibirsk (in Russian).

Zhao, X., Coe, R.S., Liu, C. & Zhou, Y., 1992. New Cambrian and Ordovician paleomagnetic poles for the North China block and their paleogeographic implications, *J. geophys. Res.,* **97,** B2, 1767–1788.

Zijderveld, J.D.A., 1993. A.C. demagnetization of rocks: analysis of results, in: *Methods in Palaeomagnetism,* pp. 254–286, eds Collinson, D.W., Creer, K.M. & Runcorn, S.K., Elsevier, New York, NY.

Zinchenko, V.N., Vasilieva, N.I. & Rudavskaya, V.A., 1993. The basal Cambrian depositions of the North-Eastern part of Siberian platform, in *Stratigrafiya fanerozoya neftegazonosnih regionov Rossii,* pp. 4–18, VNIGRI, St Petersburg (in Russian).